

Biomedical Computer Vision

Image Segmentation & Post Processing

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Virginia Tasso
virginia.tasso@polimi.it



Syllabus:

17.03

- Data dimensionality, image formats
- Image preprocessing

19.03

- Introduction to convolution
- Padding, stride, pooling

24.03

- Convolution-based operation: filtering (denoising, sharpening, edge detection)

27.03

- Introduction to segmentation: thresholding, masking, k-means...
- Post-processing: connected components, erosion, dilation...

31.03

- Introduction to Deep Learning
- Fully Connected Neural Networks, Convolutional Neural Networks

02.04

- NN training and regularization (dropout, weight decay...)

14.04

- Image segmentation with Deep Learning
- UNet

16.04

- CNN and others for other tasks (generation, denoising...)
- Review of the state of the art

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Image Segmentation

Image segmentation represents the process by which the image is subdivided into regions or objects.

The segmentation of complex images is one of the most challenging tasks in the field of image processing. The accuracy and precision of the segmentation directly determine the success or failure of subsequent computer-based analysis procedures.

Image segmentation methods will look for objects that either have some measure of homogeneity within themselves, or have some measure of contrast with the objects on their border. The homogeneity and contrast measures can include features such as gray level, color, and texture.

Image Segmentation Methods

Image segmentation techniques can be divided into four main categories:

- Histogram thresholding
- Clustering methods
- Region growing

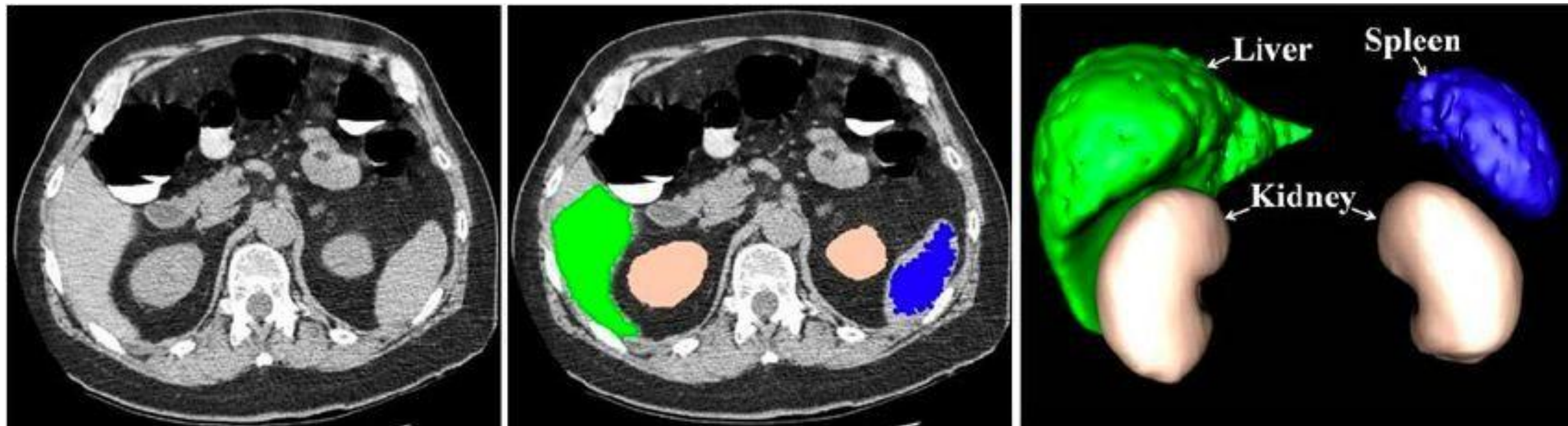
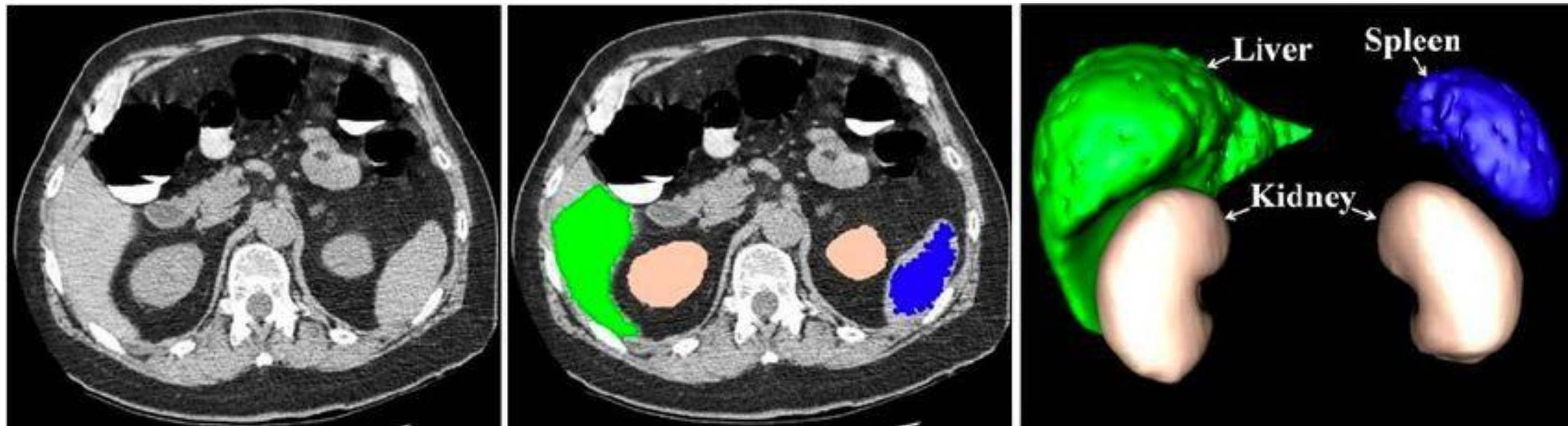


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Thresholding

Thresholding is the process of separating an image into different regions based upon its gray level distribution.

Separation of the object pixels from the background is accomplished by selecting a gray level T such as all pixels with $f(x,y) > T$ will be classified as belonging to the object.

This process reduces a multilevel image into a binary image:

$$g(x, y) = \begin{cases} G_a & f(xy) \leq T \\ G_b & f(xy) > T \end{cases}$$

Thresholding types (bimodal histogram)

Key in the selection of the threshold value is the image histogram.

When T is applicable to the entire image, we speak of **global** thresholding.

If T changes over the image, we use **variable (local, regional)** thresholding.

If T depends on the pixel coordinates, we speak of **dynamic**, or **adaptive**, thresholding.

Multilevel (multiple) thresholding can help in presence of more objects on the background

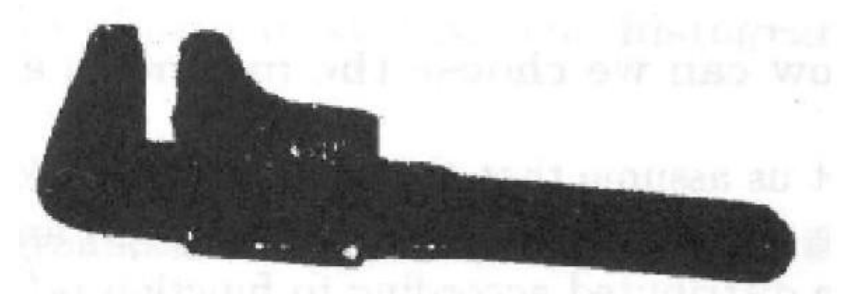
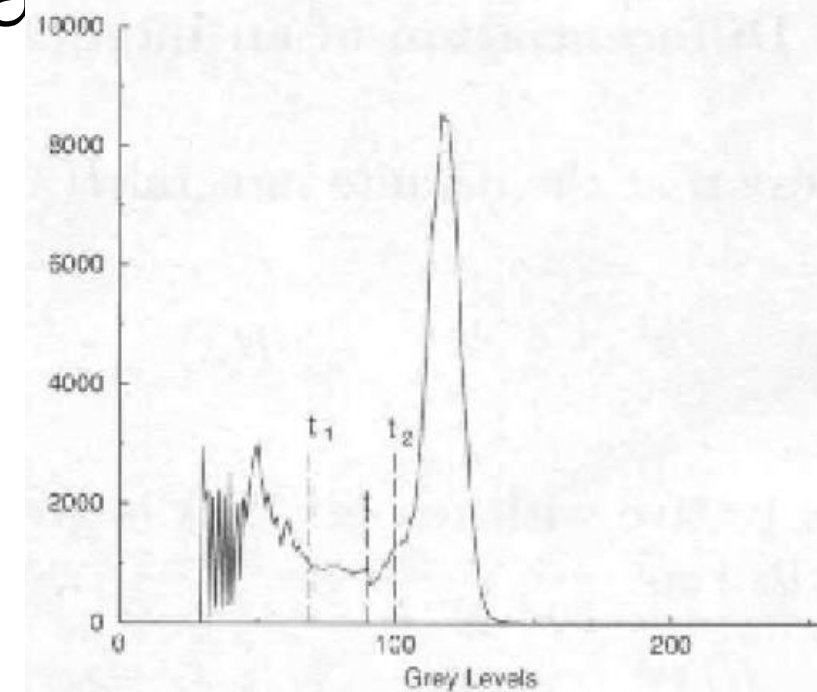
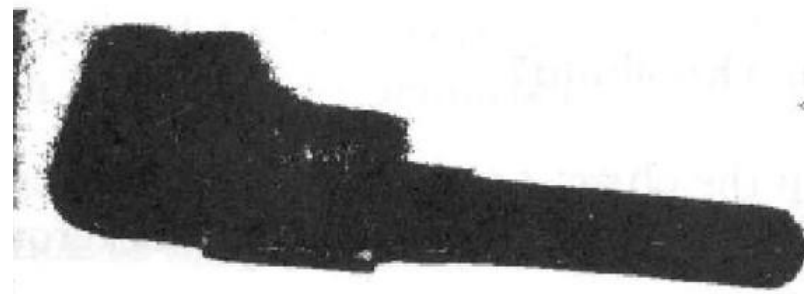
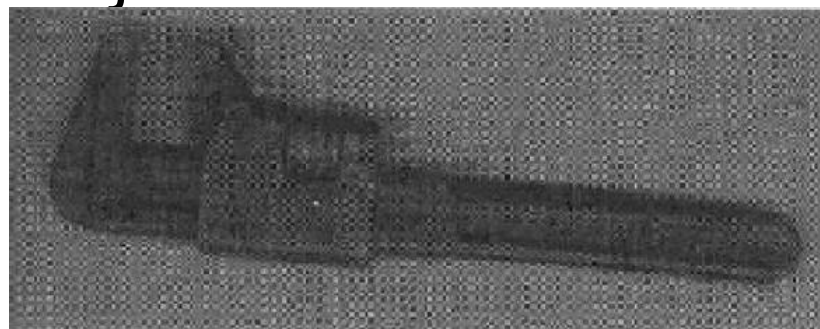
$$g(x, y) = \begin{cases} G_a & 0 \leq f(xy) < T_1 \\ G_b & T_1 \leq f(xy) < T_2 \\ G_c & T_2 \leq f(x, y) \leq G_{max} \end{cases}$$

Hysteresis Thresholding

If there is no clear valley in the histogram of an image, then there are several background pixels that have similar gray level value with object pixels and vice versa.

Hysteresis thresholding (i.e., two thresholds, one at each side of the valley) can be used in this case. Pixels above the high threshold are classified as object and below the low threshold as background.

Pixels between the low and high thresholds are classified as object only if they are adjacent to other object pixels.



Thresholding Conditions

The success of this approach is related directly to the width and depth of the valley(s) separating the histogram modes.

High object-to-background contrast
&
Good Illumination

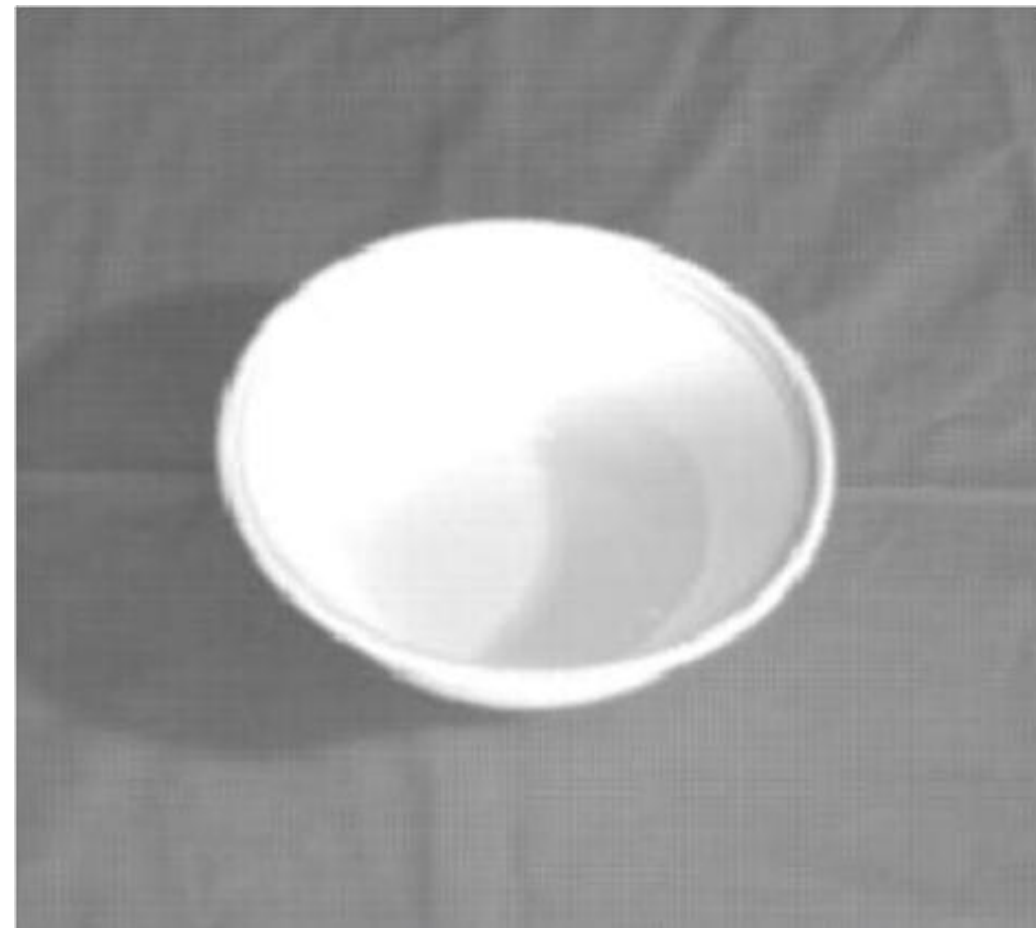
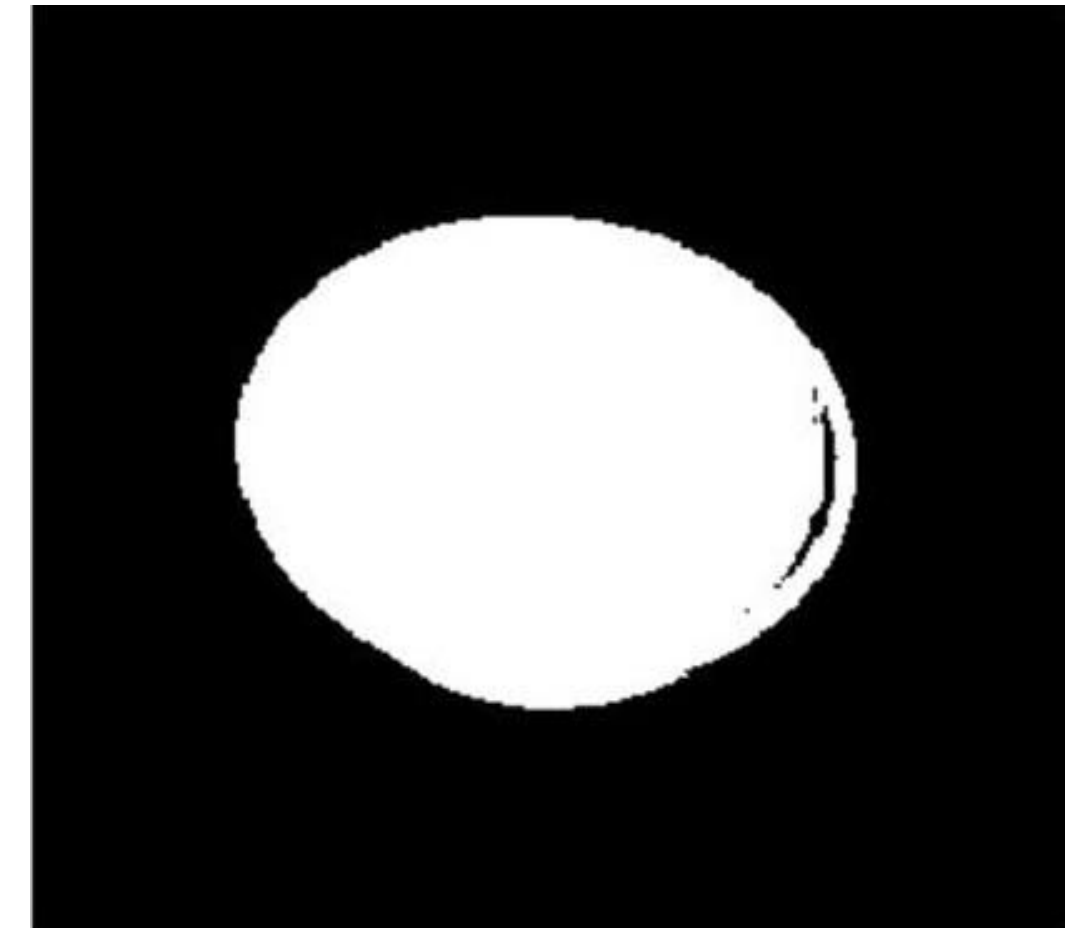


Image Histogram



After image threshold



Thresholding Conditions

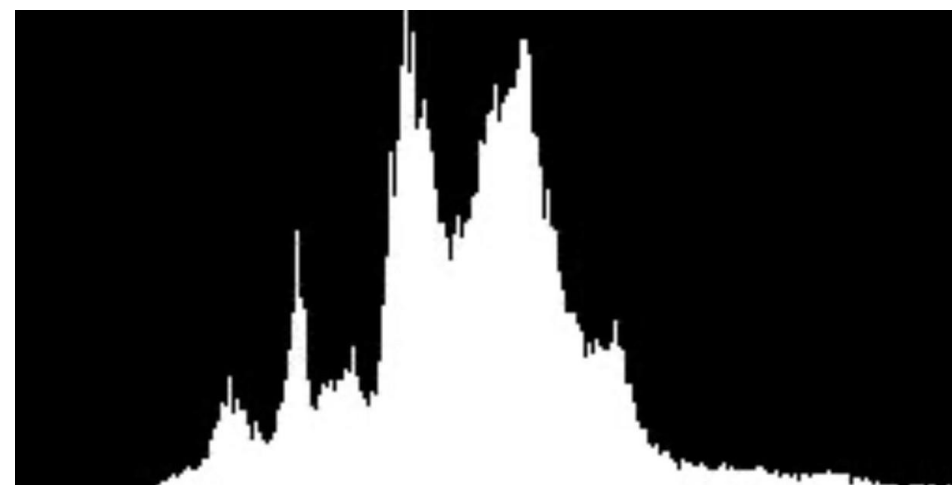
These are affected by:

- 1) the separation between peaks
- 2) the noise content (it broadens the modes)
- 3) the relative size of objects and background
- 4) the uniformity of the illumination source
- 5) the uniformity of the reflectance properties in the image

Poor object-to-background contrast
Poor Illumination



Image Histogram

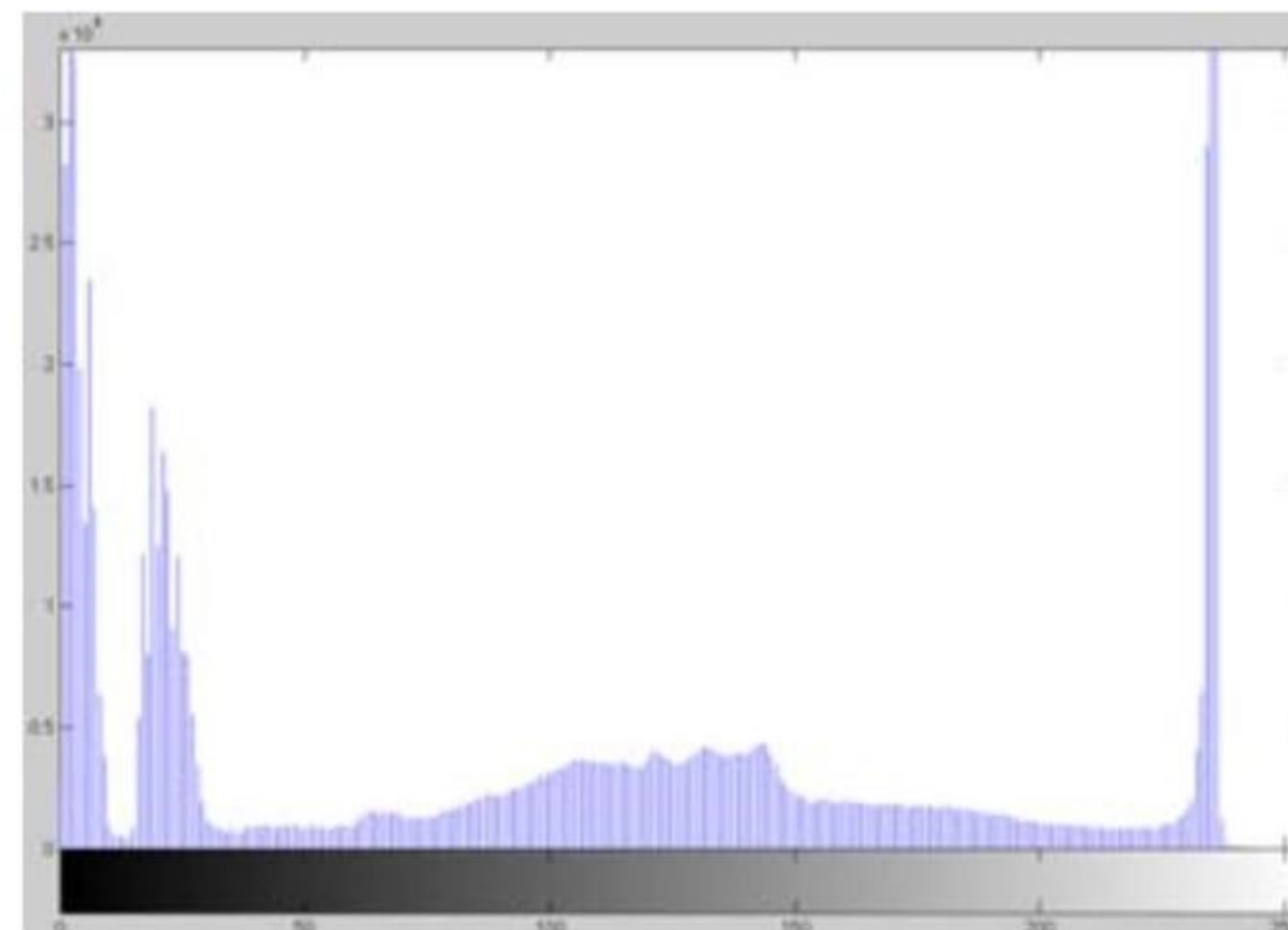


After image threshold



Thresholding Conditions

Typically, the histogram of a biomedical image is anything but bimodal, so this approach is applicable only to region of interests and the selection of the best threshold is at best a compromise between the segmentation of the different features that are present within the image. A trial and error method is usually used to select the threshold value.



Otsu's Method

This method approaches the segmentation as a classification problem. The goal is to select a threshold value that minimizes the areas of the two classes (within-class variance) while maximizing the distance (between-class variance) between them.

For every possible threshold T :

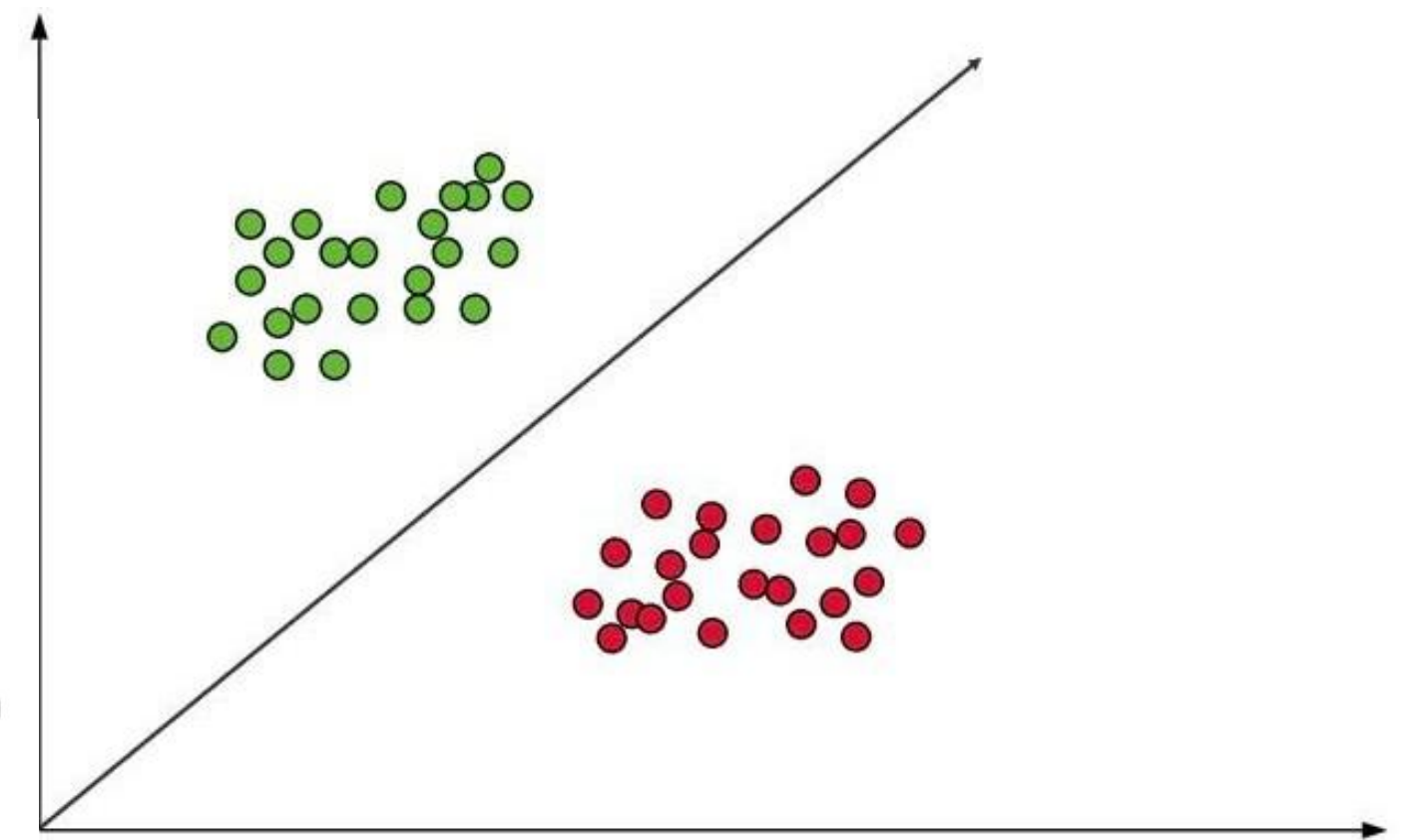
A. Calculate within group variances:

1. probability of being in group 1
probability of being in group 2
2. determine mean of group 1;
determine mean of group 2

$$q_1(T) = \sum_{i=0}^T p(i)$$

$$q_2(T) = \sum_{i=T+1}^{max} p(i)$$

Decision Function



Otsu's Method

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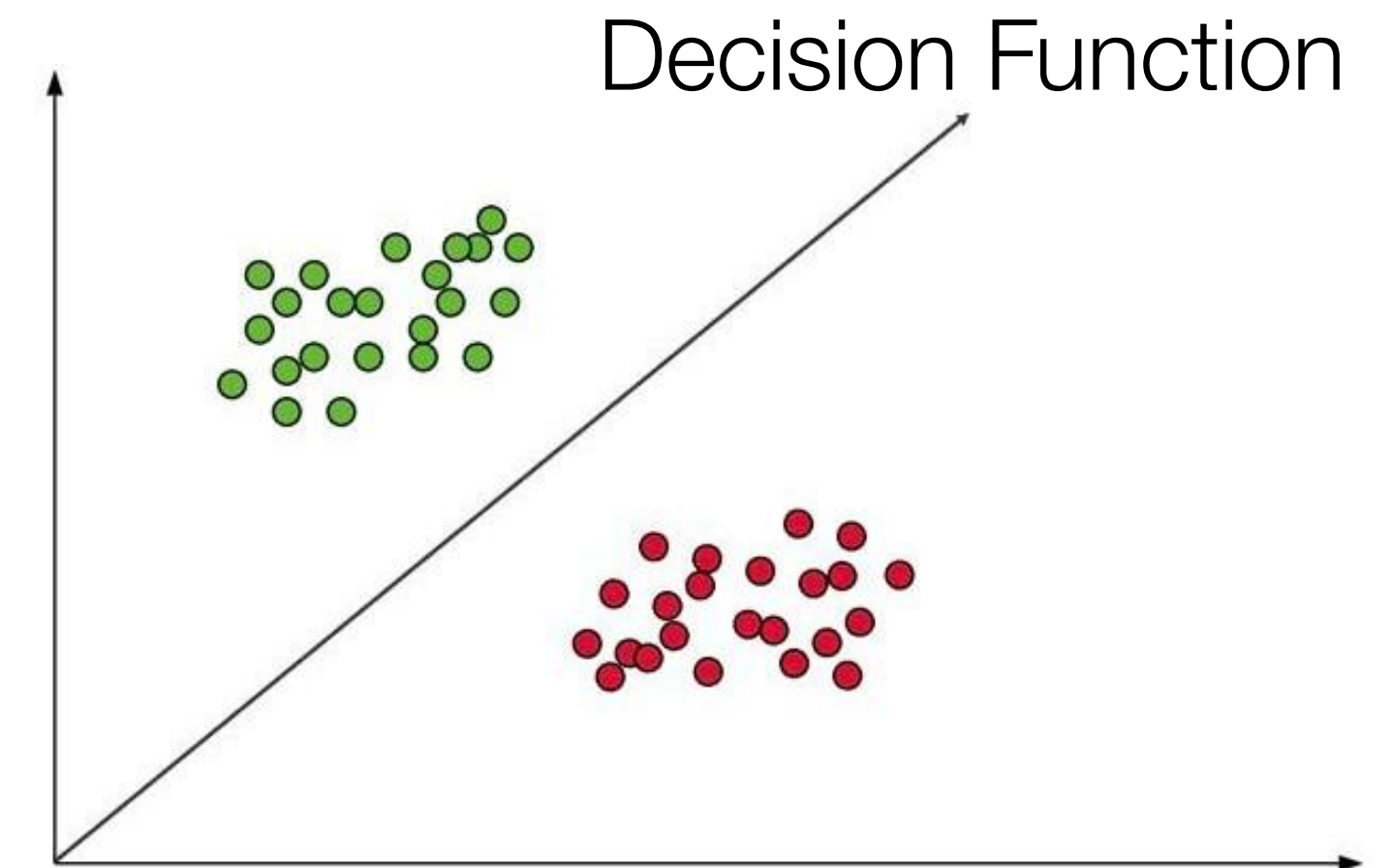
For every possible threshold T :

A. Calculate within group variances:

3. calculate the variance for group 1

calculate the variance for group 2

$$\mu_1(T) = \sum_{i=0}^T i \frac{p(i)}{q_1(t)} \quad \sigma_1^2(T) = \sum_{i=0}^T [i - \mu_1(t)]^2 \frac{p(i)}{q_1(t)}$$
$$\mu_2(T) = \sum_{i=T+1}^{\max} i \frac{p(i)}{q_2(t)} \quad \sigma_2^2(T) = \sum_{i=T+1}^{\max} [i - \mu_2(t)]^2 \frac{p(i)}{q_2(t)}$$



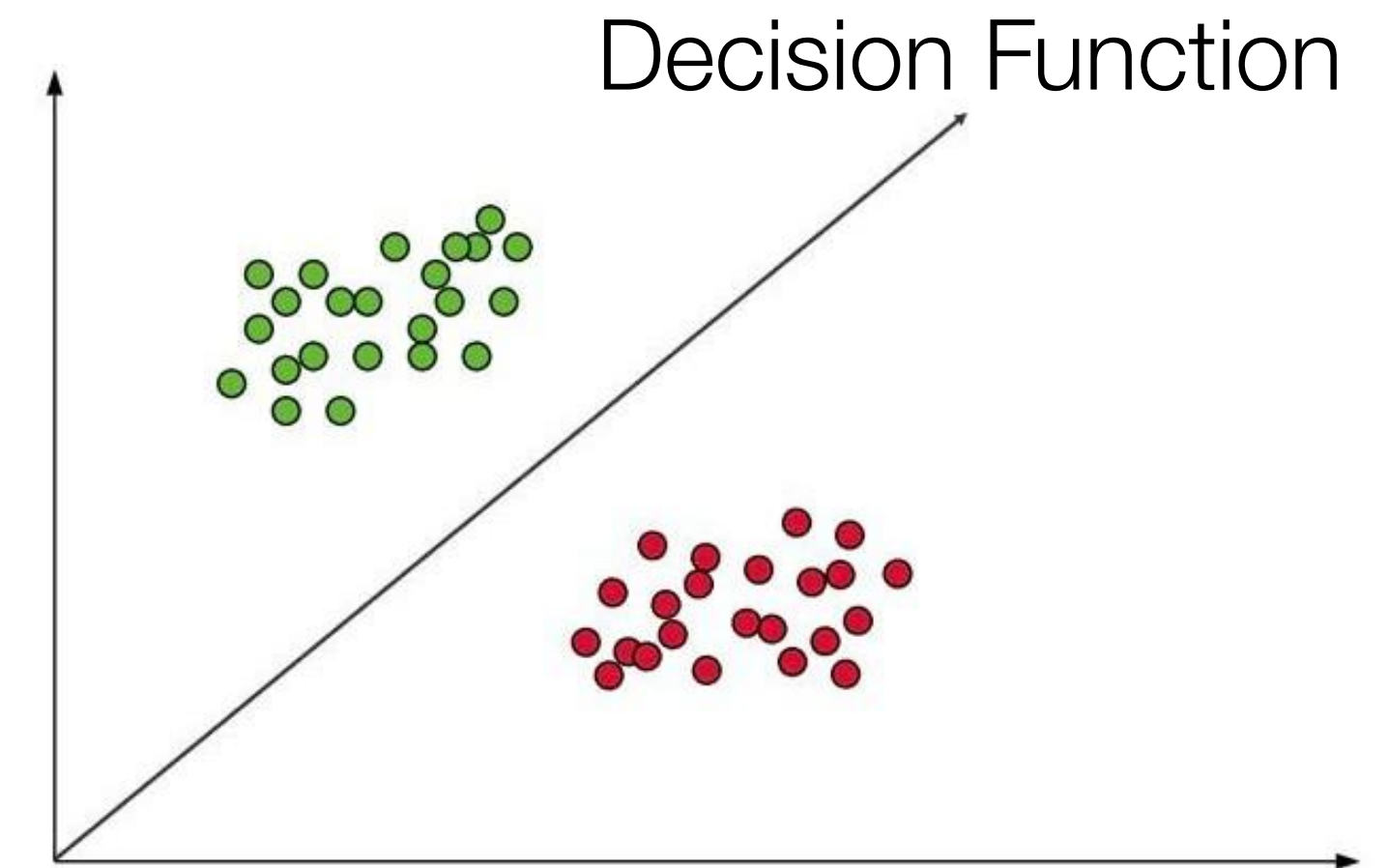
Otsu's Method

This method approaches the segmentation as a classification problem. The goal is to select a threshold value that minimizes the areas of the two classes (within-class variance) while maximizing the distance (between-class variance) between them.

For every possible threshold T :

- A. Calculate within group variances:
- 4. calculate weighted sum of group variances

$$\sigma_W^2(T) = q_1(T)\sigma_1^2(T) + q_2(T)\sigma_2^2(T)$$

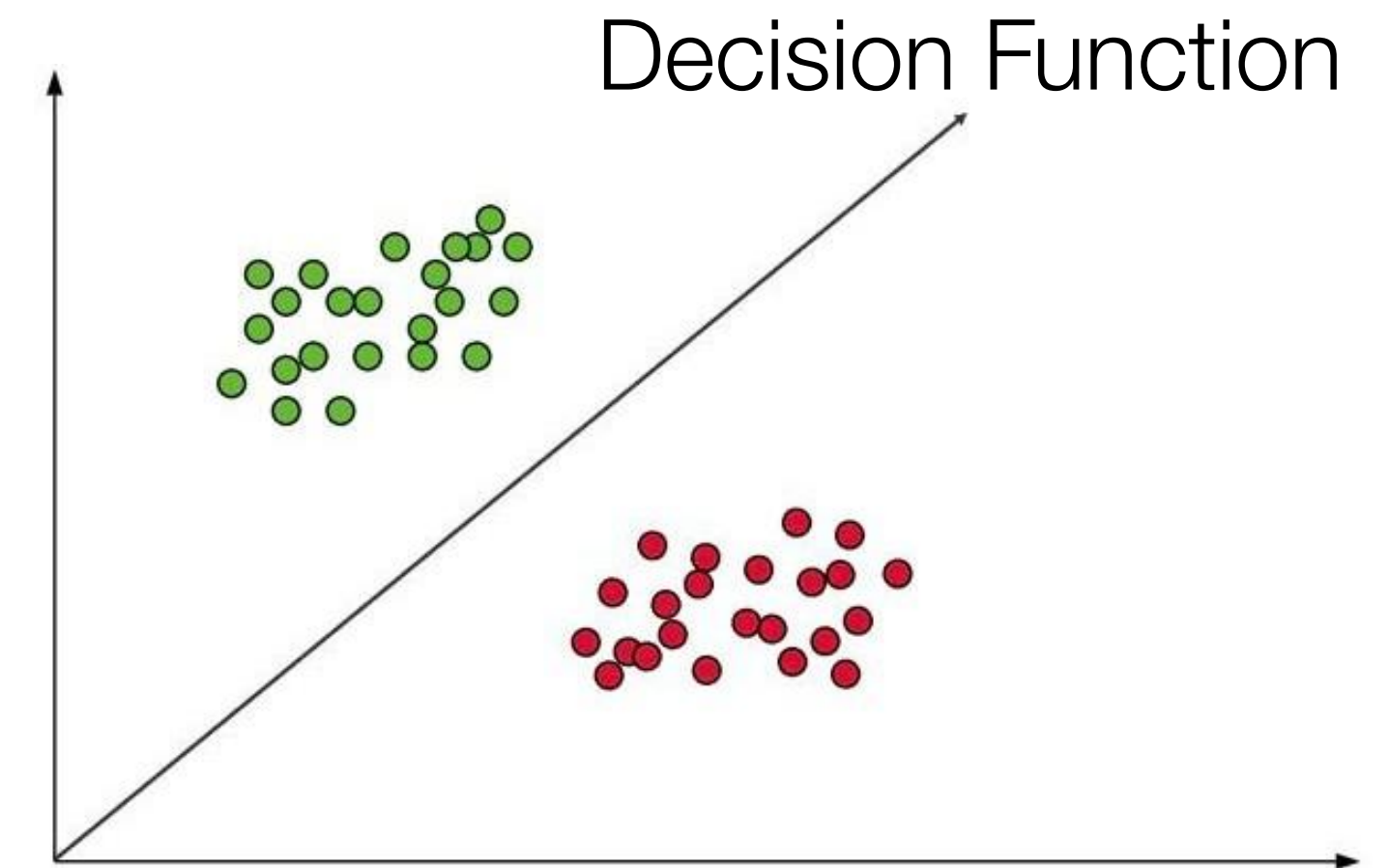


Otsu's Method

This method approaches the segmentation as a classification problem. The goal is to select a threshold value that minimizes the areas of the two classes (within-class variance) while maximizing the distance (between-class variance) between them.

For every possible threshold T :
B. Remember which t gave rise to minimum:

$$\min\{\sigma_w^2(T) \mid 0 \leq T \leq \max\}$$



Otsu's Method - Drawbacks

The method assumes that the histogram of the image is bimodal (i.e., two classes).

The method breaks down when the two classes are very unequal (i.e., the classes have very different sizes)

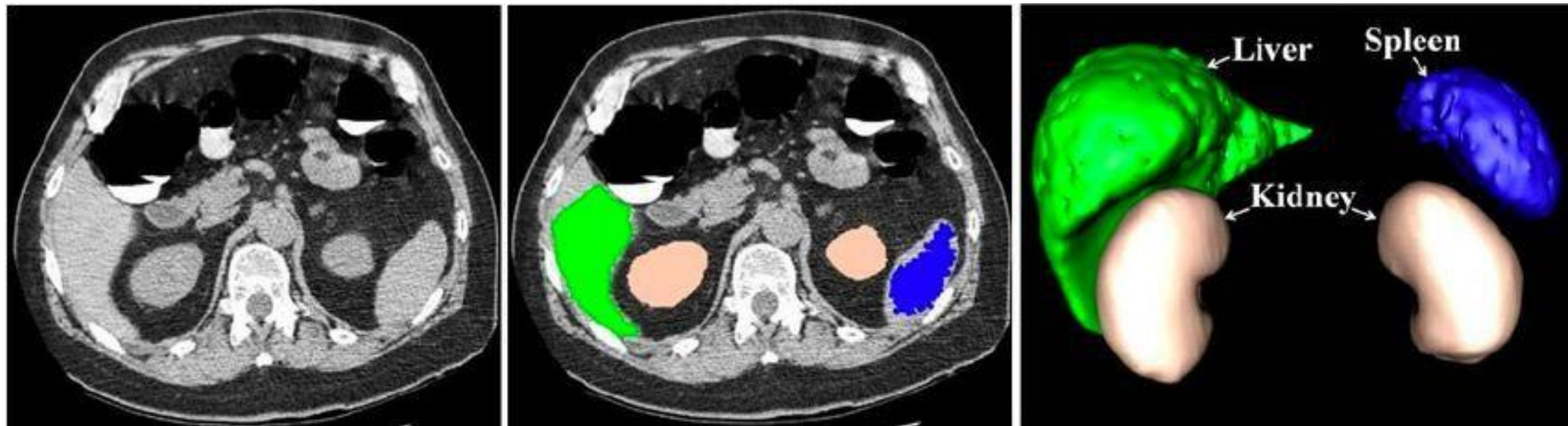
The correct maximum is not necessary the global one.

The method does not work well with variable illumination.

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K-Means Clustering

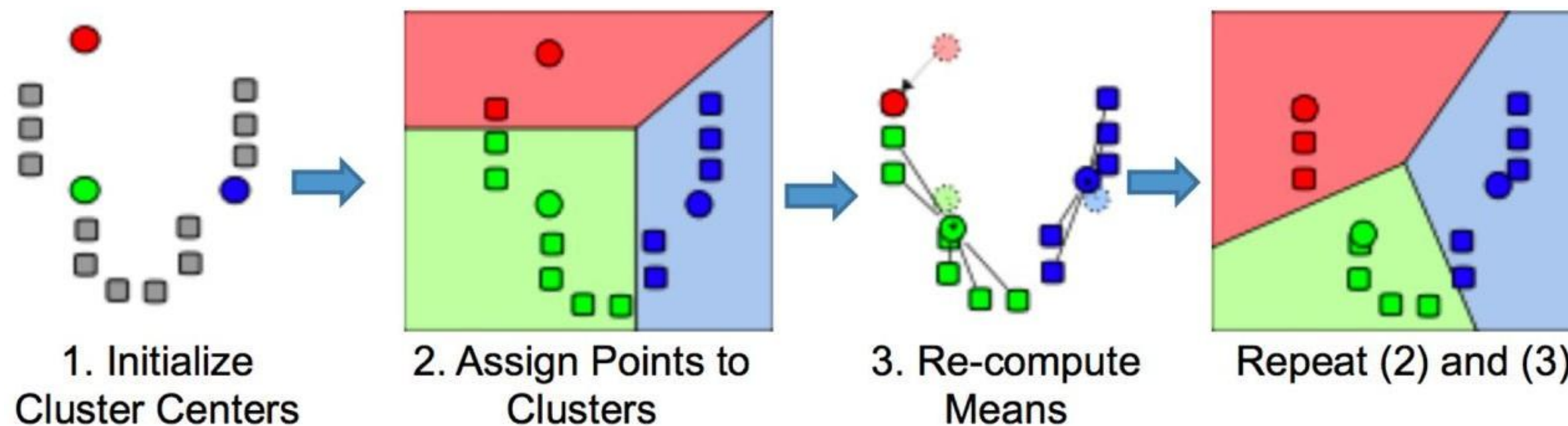
In statistics and machine learning, K-means clustering is a method of cluster analysis which aims to partition n observations into K clusters in which each observation belongs to the cluster with the nearest mean.

- Patterns in a cluster are similar to others in the same cluster, and different from patterns in the other clusters
- Each cluster represents an abstract concept (in images, clusters represent regions with similar intensity)
- A form of unsupervised learning (No apriori information about the groups)
- Clustering = the process of partitioning a set of pattern vectors into subsets called clusters.
- K = number of clusters (must be known in advance).
- Not an exhaustive search so it may not find the globally optimal solution.

K-Means Clustering

Form K-means clusters from a set of nD feature vectors.

1. Set $i_c = 1$ (iteration count).
2. Choose randomly a set of K means $m1(1), m2(1), \dots, mK(1)$.
3. For each vector x_i compute $D(x_i, m_j(i_c))$ for each $j=1, \dots, K$.
4. Assign x_i to the cluster C_j with the nearest mean.
5. $i_c = i_c + 1$; update the means to get a new set $m1(i_c), m2(i_c), \dots, mK(i_c)$.
6. Repeat 3..5 until $C_j(i_c + 1) = C_j(i_c)$ for all j .



K-Means Clustering

Strength:

- Relatively efficient: $O(tkn)$, where n is # objects, k is # clusters, and t is # iterations.
Normally, $k, t \ll n$.
- Often terminates at a local optimum.
- K-Means readily generalizes to: arbitrary number of classes (K) and can easily be extended to many, many features (i.e., feature vectors instead of only gray values/ higher dimensions)

Weakness:

- Applicable only when mean is defined, then what about categorical data? kmedoids
- Need to specify k , the number of clusters, in advance (Elbow method or Silhouette score)
- Unable to handle noisy data and outliers
- Not suitable to discover clusters with non-convex shapes

K-Means Clustering

- **Weakness:**

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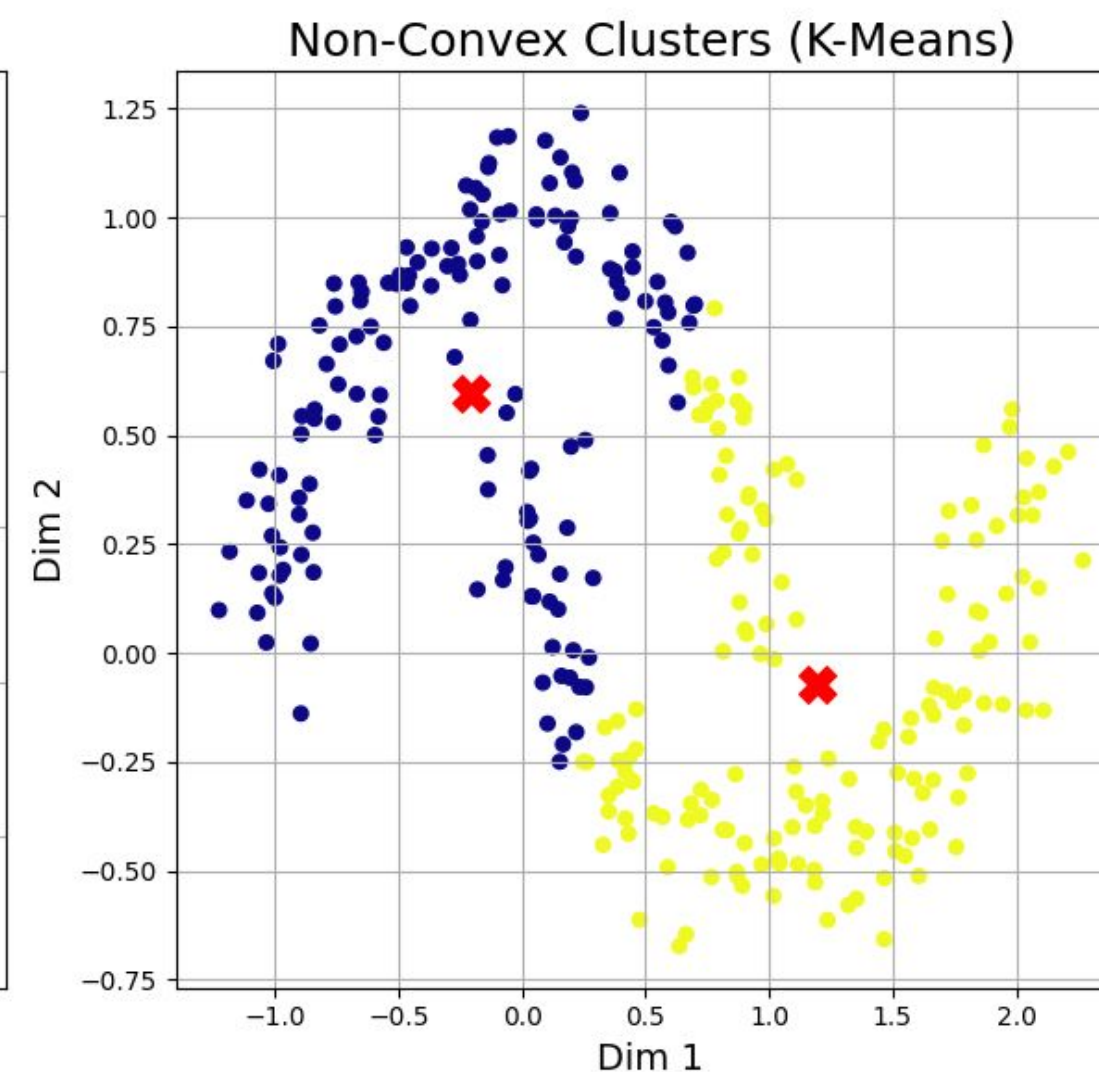
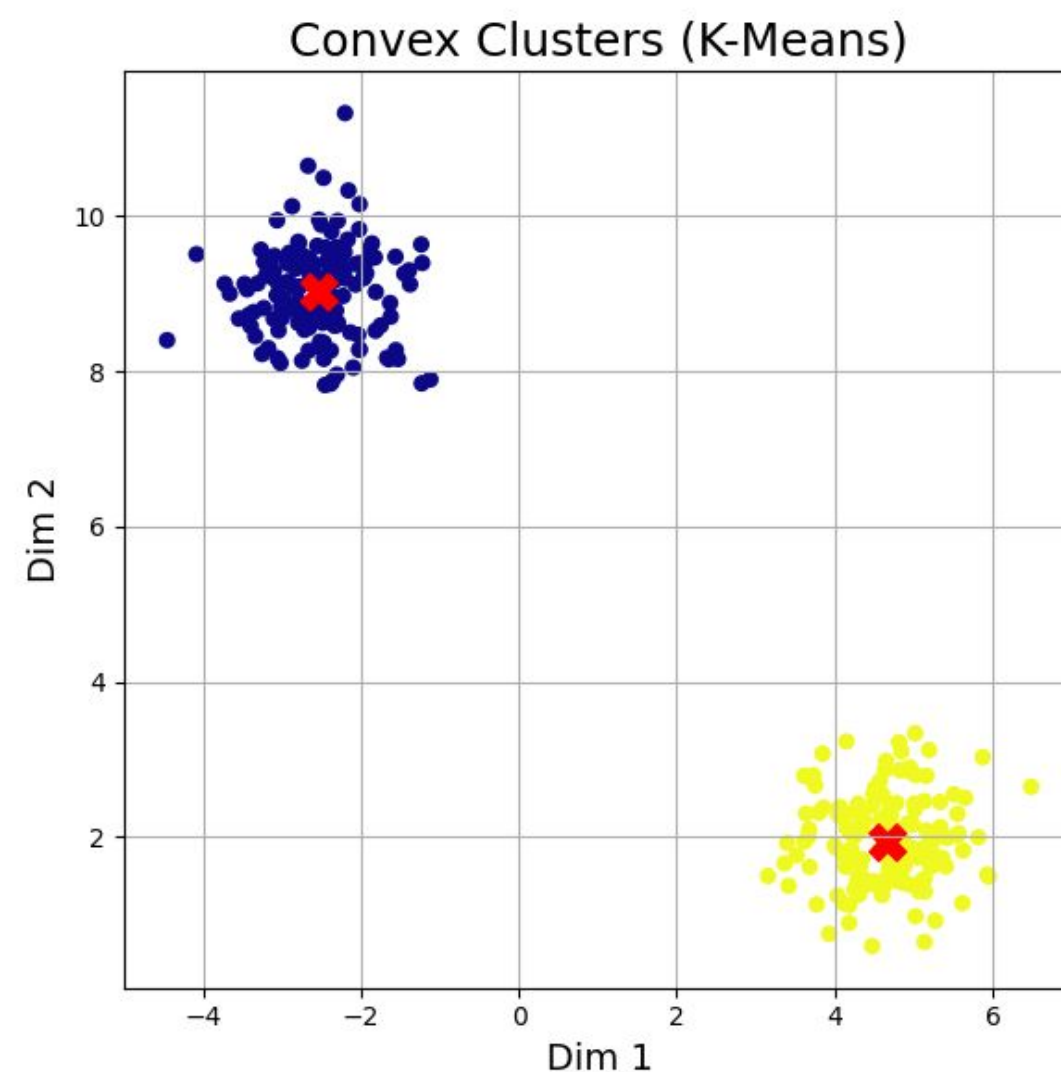
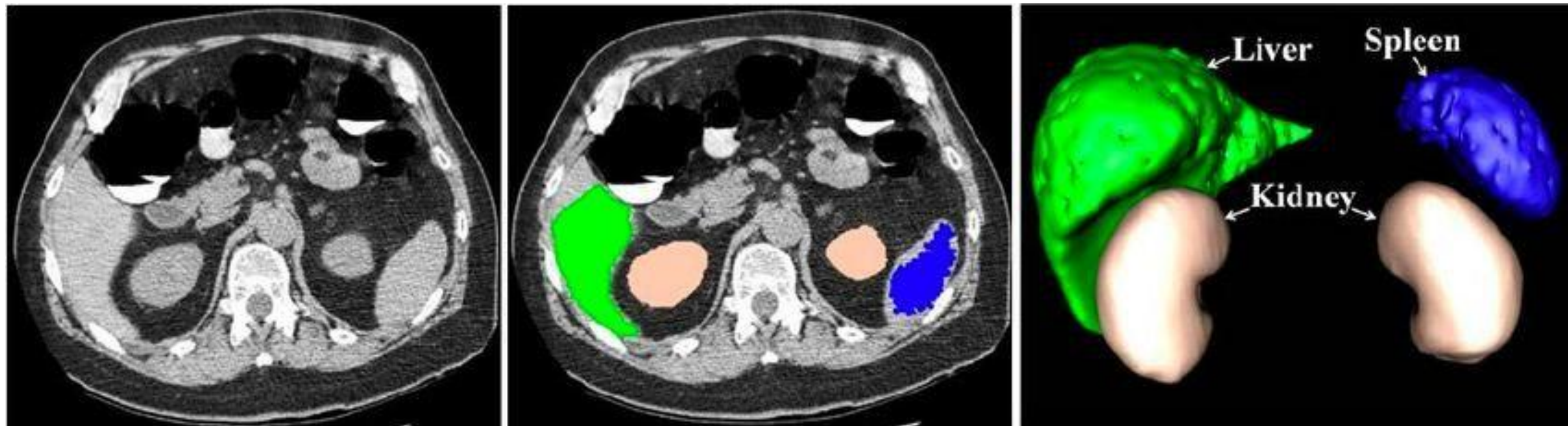


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Region Growing

This category includes segmentation strategies based on local uniformity, instead then local differences. Pixels, or sub-regions, are merged in larger regions, based on predetermined growing criteria. It is an iterative method used to extract similar parts of an image, exploiting the fact that pixels which are close together have similar gray values.

1. Start with a single pixel (seed) and add new pixels slowly

How do we choose the seed(s) in practice?

- It depends on the nature of the problem.
- If targets need to be detected using infrared images for example, choose the brightest pixel(s).
- Without a-priori knowledge, compute the histogram and choose the gray-level values corresponding to the strongest peaks

Region Growing

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Region-growing approaches exploit the fact that pixels which are close together have similar gray values.

2. Check the neighboring pixels and add them to the region if they are similar to the seed. How do we choose the similarity criteria (predicate)?

- The homogeneity predicate can be based on any characteristic of the regions in the image such as:

- average intensity
- variance
- color
- texture

Region Growing

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3. Repeat step 2 for each of the newly added pixels. Stop if no more pixels can be added

Region Growing- General Rules

The definition of criteria depends on the specific problem and on the type of image.

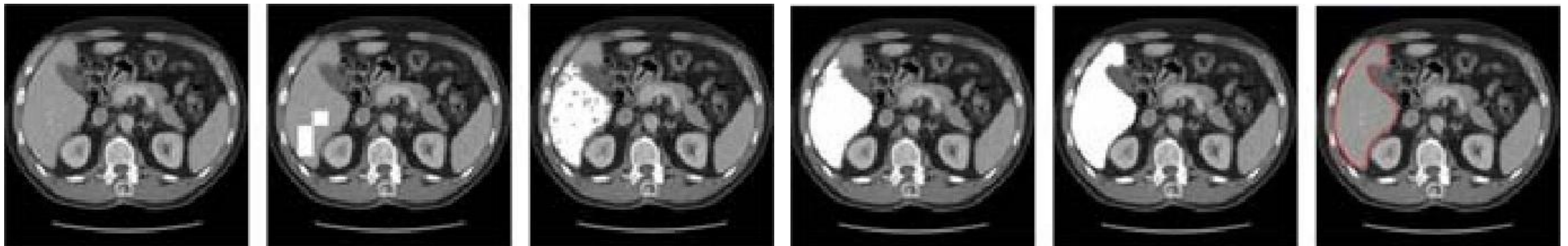
- Each region should be uniform, and homogeneous with respect to some characteristic: Color, gray scale, texture, motion
- Region interiors should be simple without many small holes
- Adjacent regions should have significantly different values/characteristics
- Boundaries of the regions should be smooth

Also, a stop rule needs to be defined: for example, when there are no new connected pixels satisfying the inclusion criteria.

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Region-growing approaches exploit the fact that pixels which are close together have similar gray values.



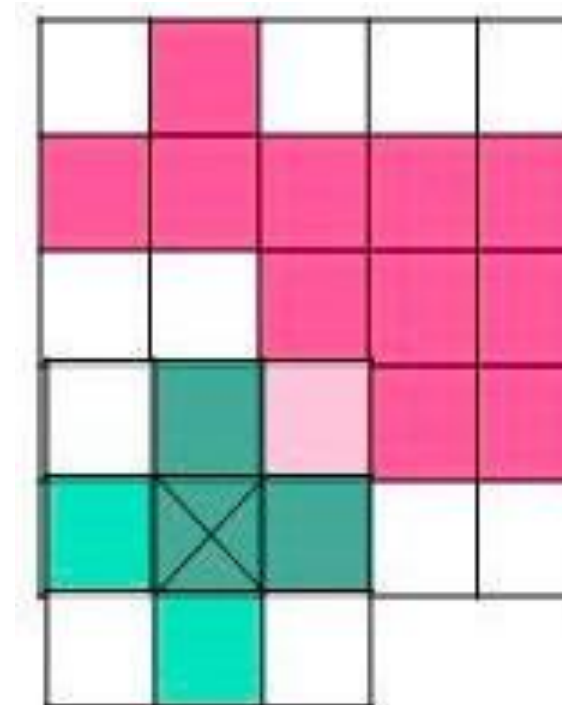
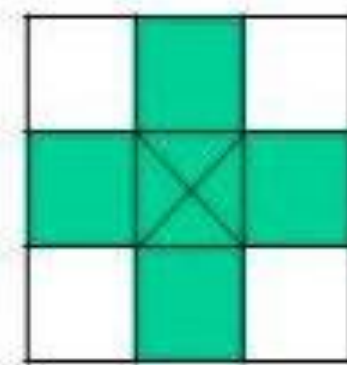
Morphological Processing

Morphological image processing is an area of nonlinear image processing that focuses on the geometrical structure of objects within an image A .

It is generally performed on binary images, but the concept can be extended also to gray scale images.

The operation of morphological processing is controlled by a structuring element, which acts as a brush or a rubber, determining how the shape of image objects is modified.

Structuring element

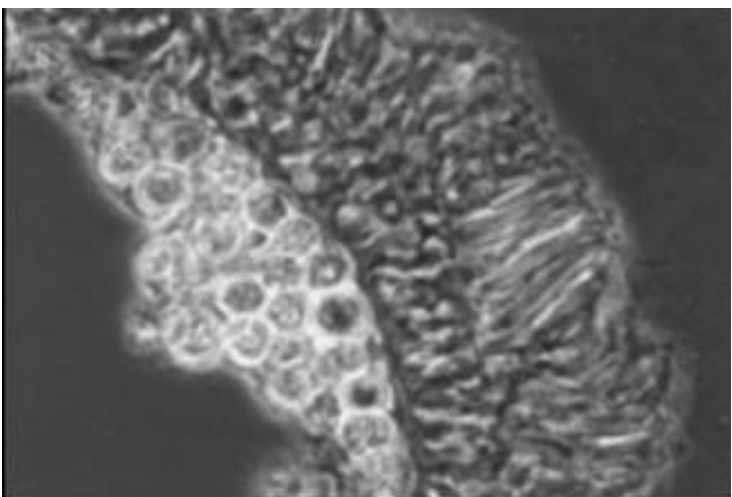


Morphological Processing - Dilation

Dilation grows or thickens objects in an image A in a way that is controlled by a shape referred to as structuring element B . For each position of B origin, it checks where B overlaps at least 1-valued pixel in A . The output will have 1 at each location of the B origin where this happens.

With a dilation operation, all the '1' pixels in the original image will be retained, any boundaries will be expanded, and small holes will be filled

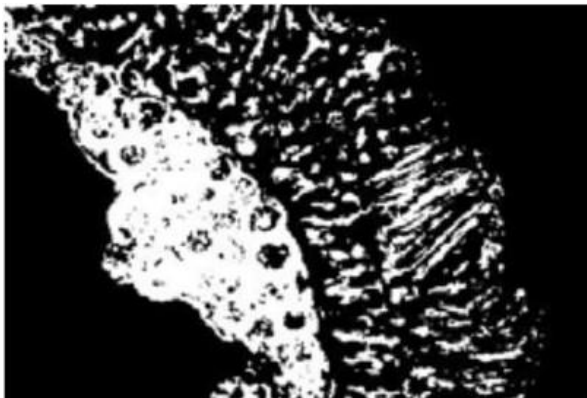
Morphological Processing - Dilation



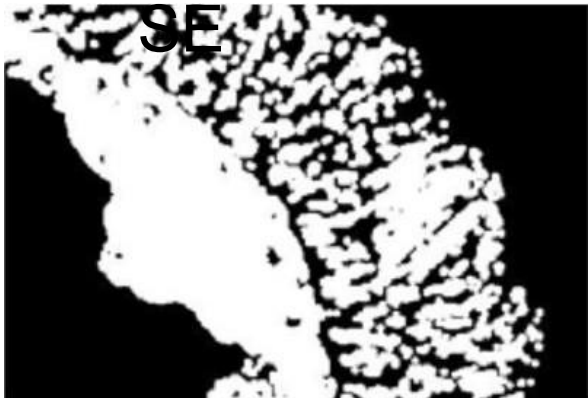
SE - Shape

SE - Size

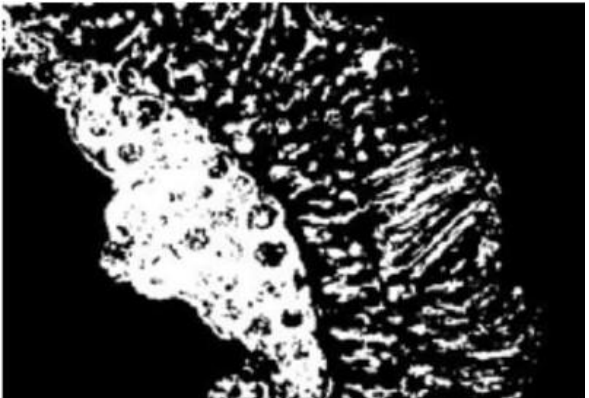
Threshold



Circular



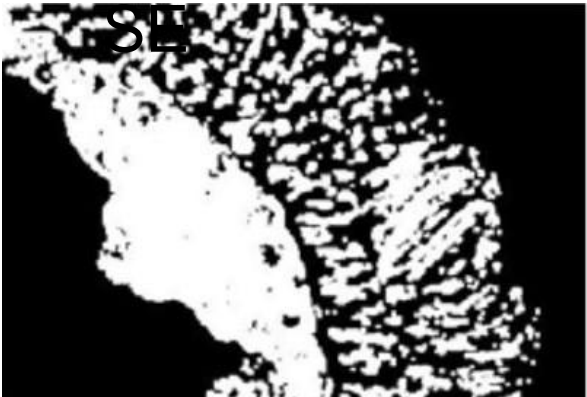
Threshold



Circular SE 3



Squared



Cross Shape



Circular SE 7



Circular SE 11



Morphological Processing - Erosion

Erosion shrinks or thins objects in a binary image, in a way that is controlled by the structuring element B . The SE is translated throughout the domain of the image to see if it fits entirely within the foreground: the output will be 1 at each location of the origin of the SE where this happens.

With an erosion operation, the only remaining pixels are those that coincide to the origin of the structuring element where it is all contained in the object

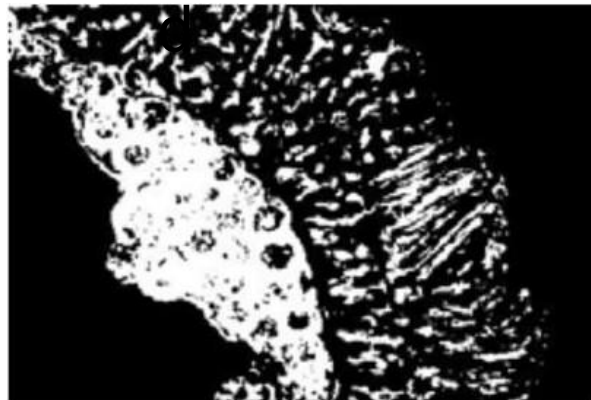
The erosion of object A by the structuring element B can be also viewed as the intersection of the two translated objects. The eroded object has been reduced in size in comparison to the original object A .

Erosion reduces the size of outward bumps, while enlarging the size of inward bumps. As the size of B increases, the size of A decreases.

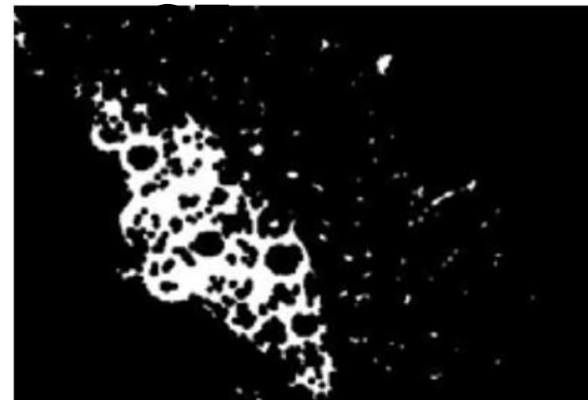
Morphological Processing - Erosion

SE - Shape

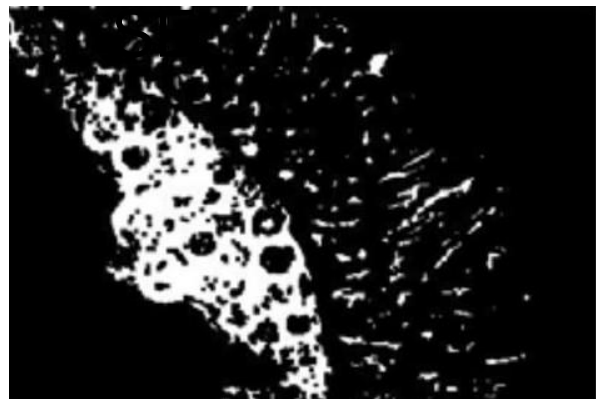
Threshol



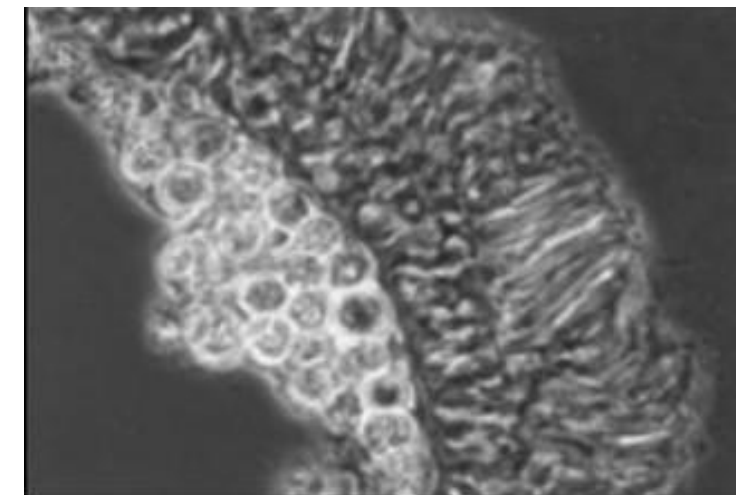
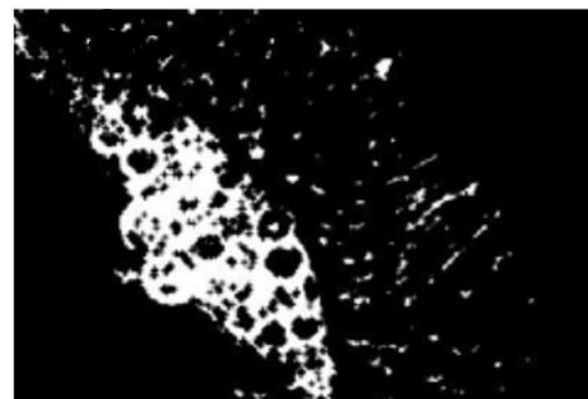
Circular



Squared



Cross Shape



Morphological Processing - Opening

The combination of dilation and erosion provides a means of maintaining the relative sizes of objects.

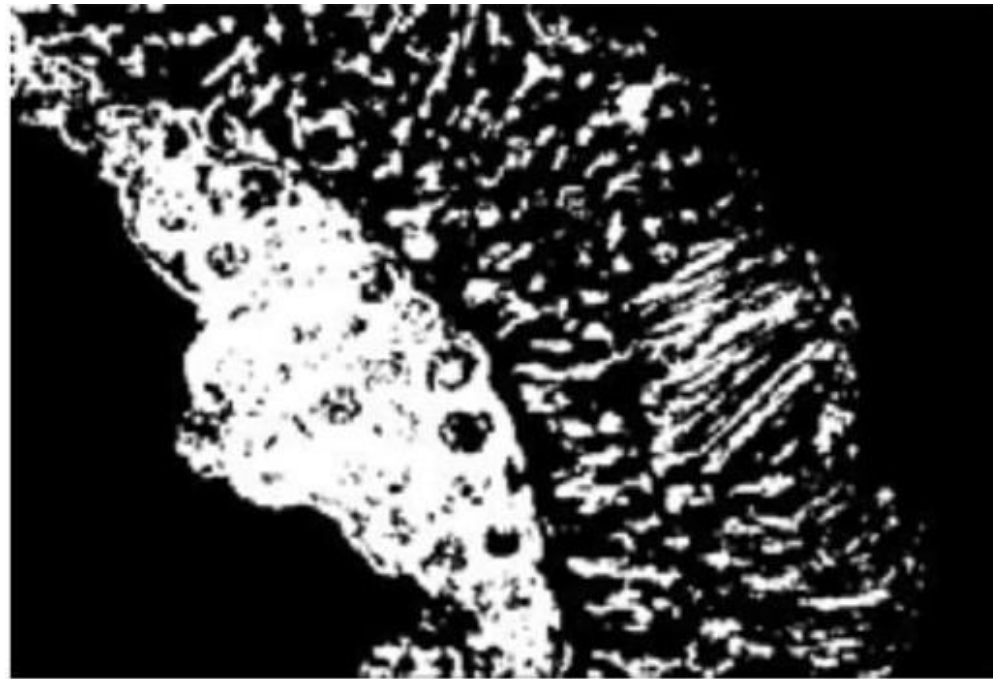
Opening:

- Erosion followed by dilation
- Remove all pixels in a region too small to contain the SE
- Disc shaped probe
- Eliminate all convex regions and isolated points
- In this case the SE is often called a probe, as it is probing the image looking for small objects to filter out of the image

The output image tends to take a shape similar to the structuring element itself

Morphological Processing - Opening

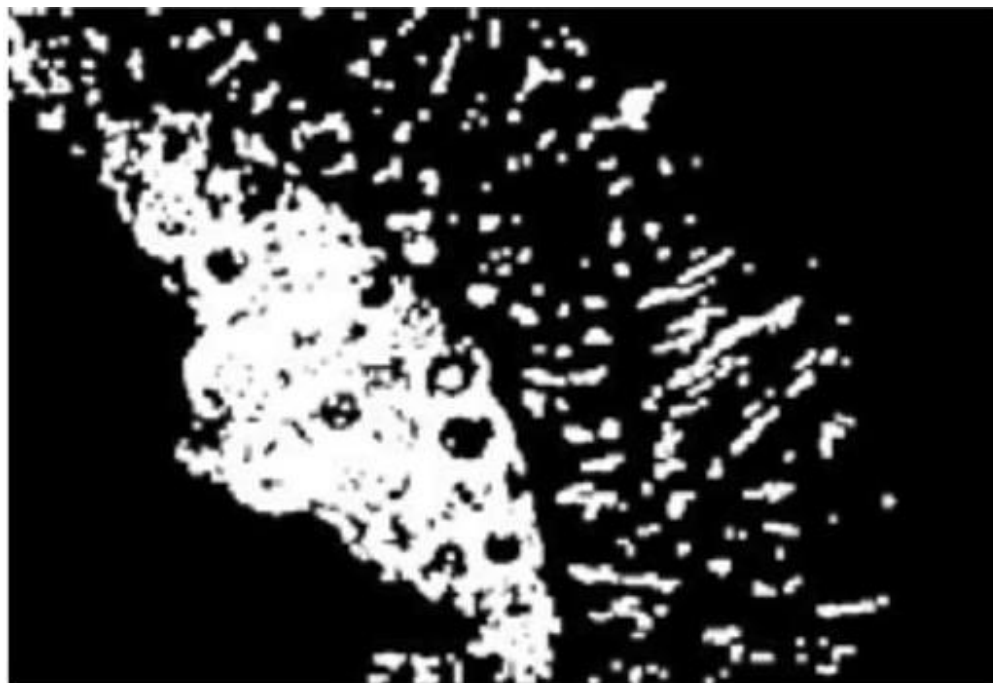
Threshold



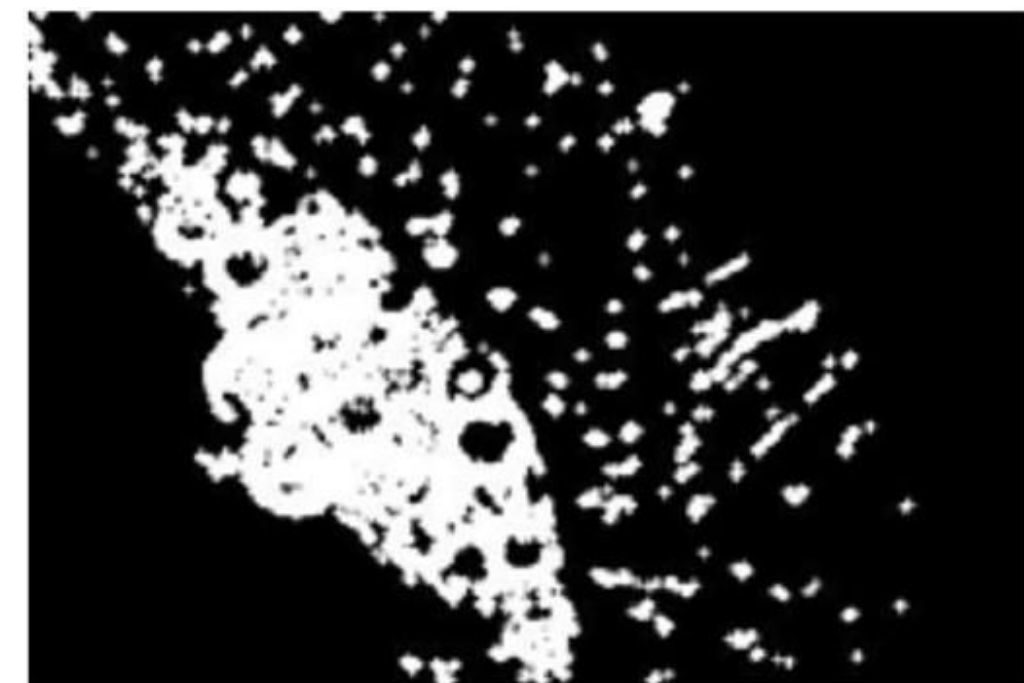
Circular SE



Squared SE



Cross Shape SE



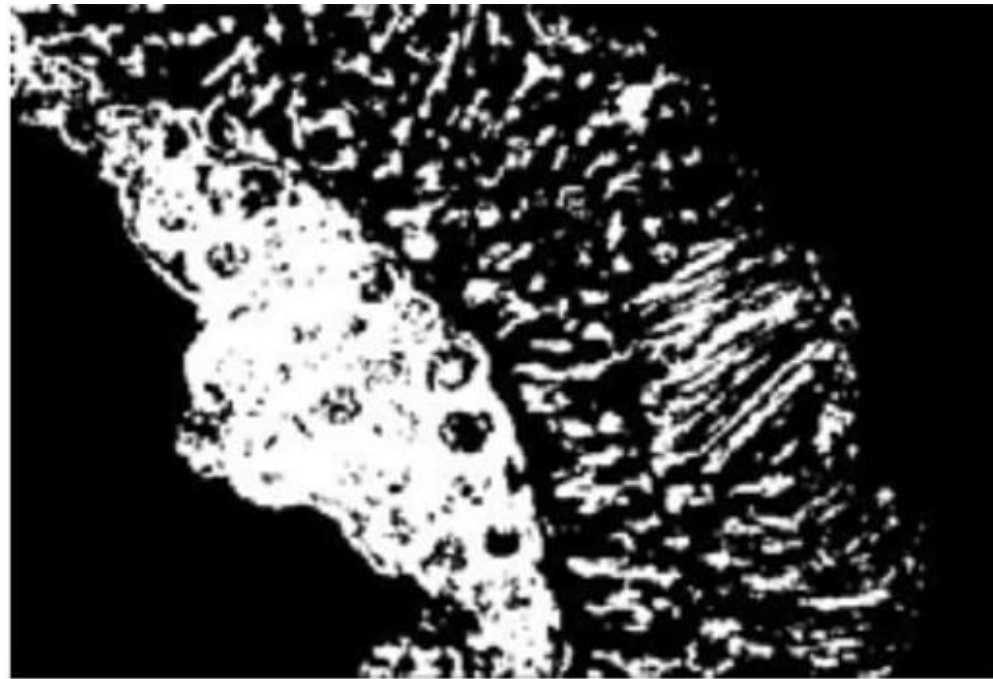
Morphological Processing - Closing

Closing:

- Opposite of opening
- Dilation followed by erosion
- Fill holes smaller than the structuring element
- It will connect small, adjacent objects
- Closing tends to “close up” or “fill in” objects

Morphological Processing - Closing

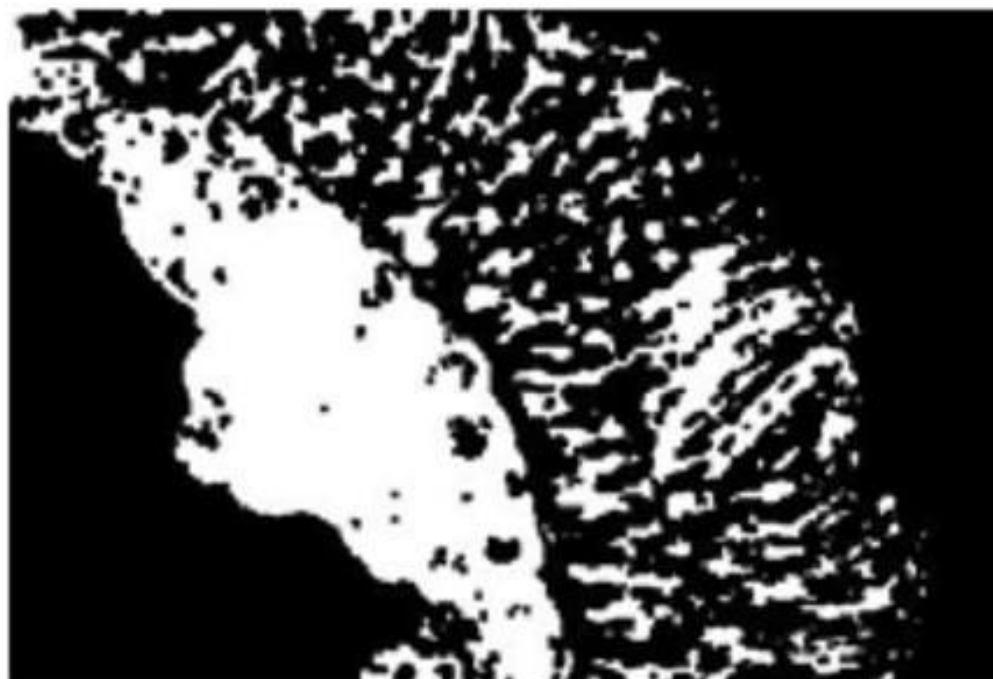
Threshold



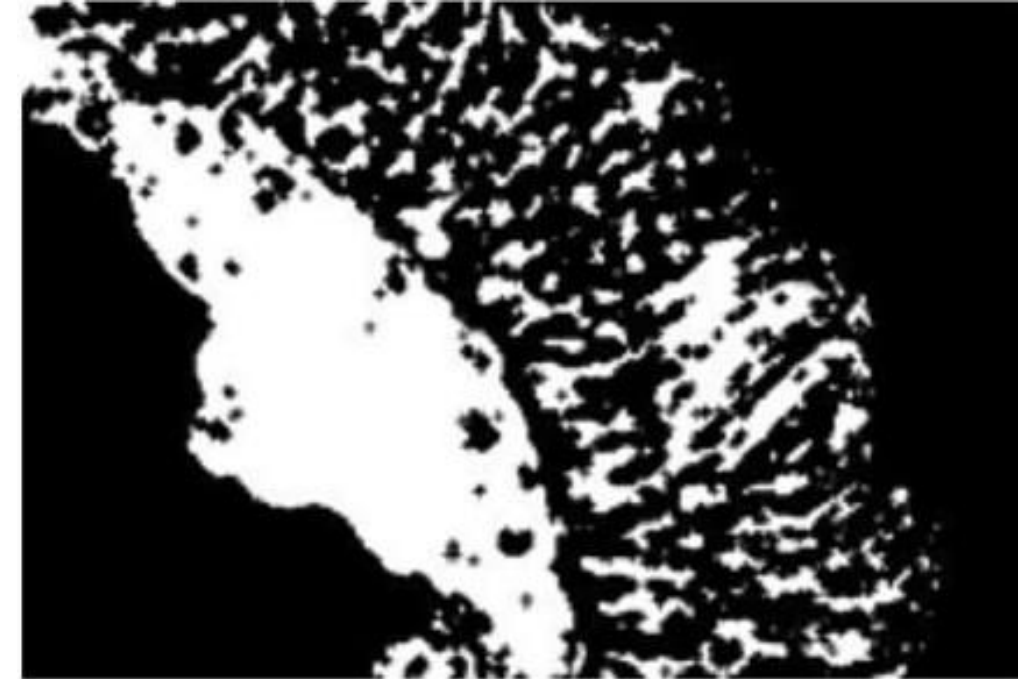
Circular SE



Squared SE

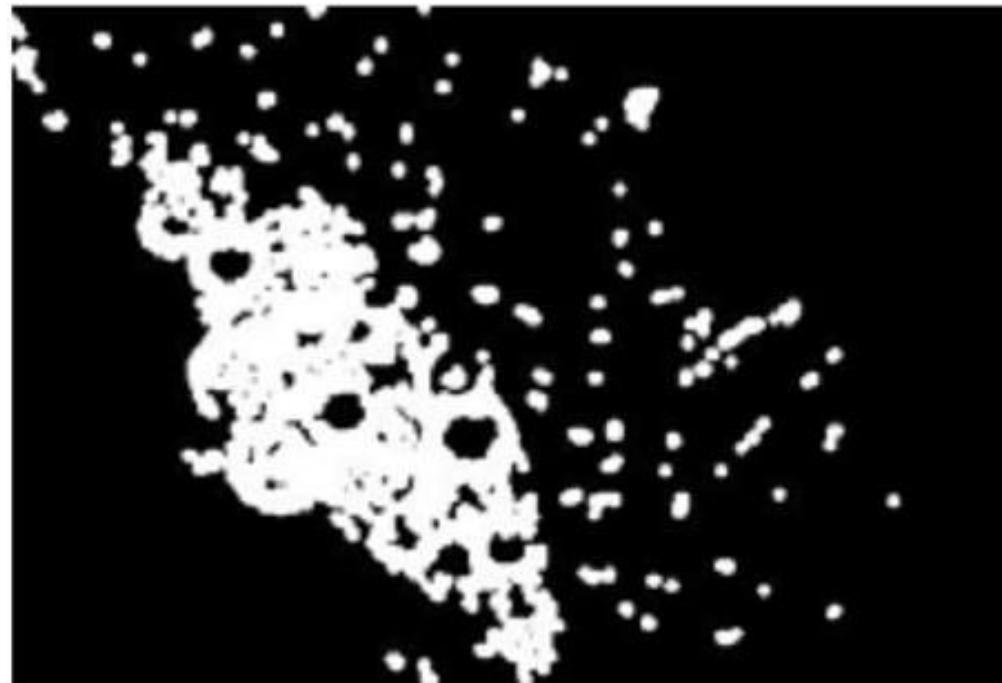


Cross Shape SE



Morphological Processing - Opening vs. Closing

Opening Circular SE 5



Closing Circular SE 5



Opening Circular SE 13



Closing Circular SE 13



Morphological Processing - Opening vs. Closing

The order of application makes the result different, as intuitively expected.

Combination of open and close can be used to remove salt or pepper noise from a binary image, with closing removing the pepper and opening removing the salt.

If both salt-and-pepper noise is present, the sequential use of closing and opening is performed:

$$\text{OPENCLOSE}(A,B) = (A_B)^B$$

$$\text{CLOSEOPEN}(A,B) = (A^B)_B$$

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